Conditionals and Truth Tables

Introduction to Logic (CS), WBCS030-05, Week 3

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Mon 14 Sep / Wed 16 Sep, 2026

Today's Lecture

1. Overview
2. Proof practice
3. Conversational implicature
4. Proofs with (bi)conditionals
5. Truth tables, tautologies and logical truths
6. Concluding

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Summary of Last Week

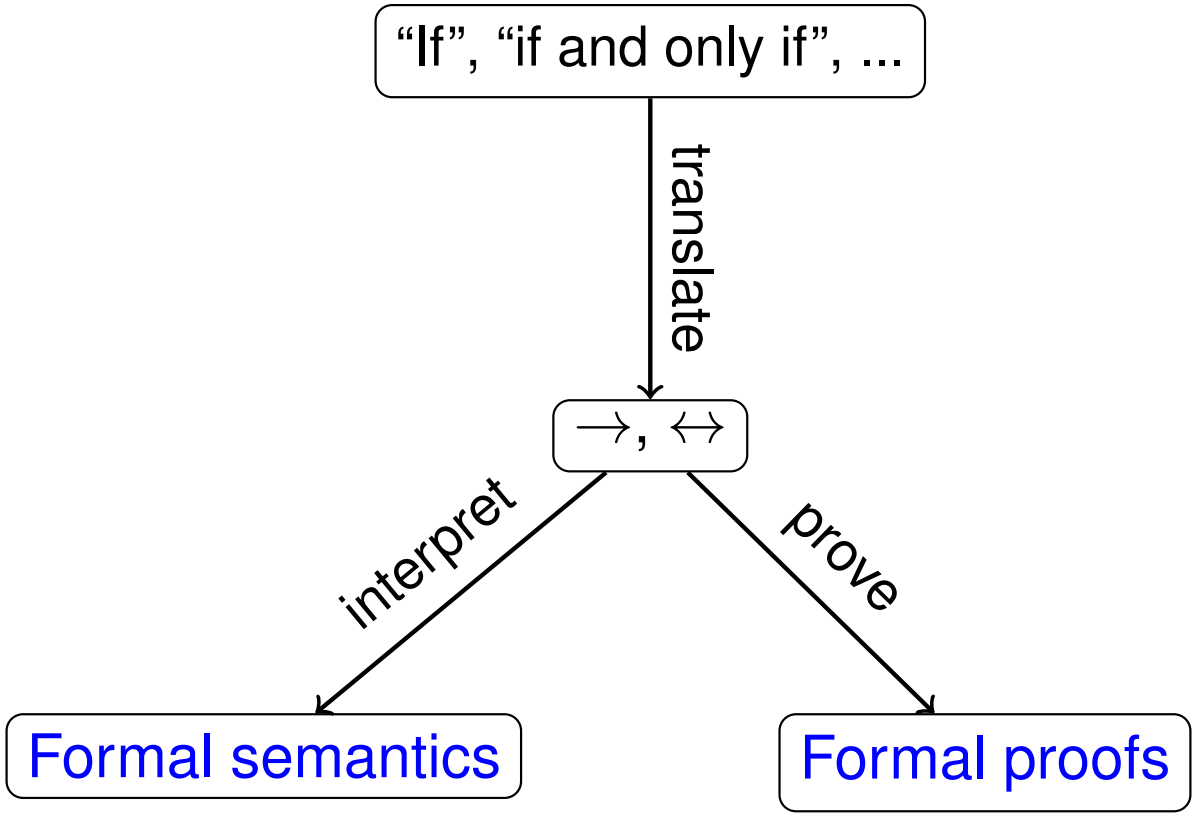
Language of first-order logic (FOL)

- Atomic sentences ✓
- Boolean connectives ✓
- Conditionals (just started)
- Quantifiers

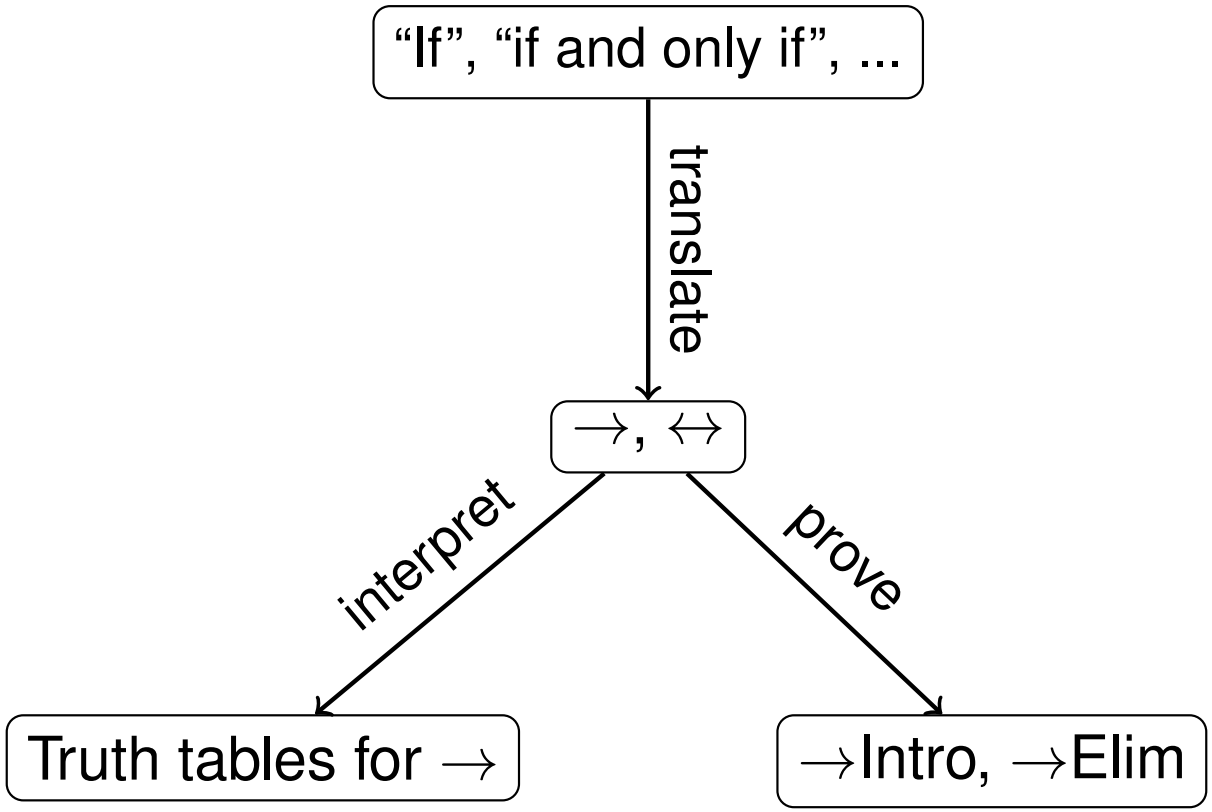
This week:

- Proof rules with conditionals and biconditionals
- Truth table semantics, tautologies, and (logical) consequence
- Conversational implicature
- Second lecture is on Wednesday!

Overview


“If”, “if and only if”, ...
translate
→, ↔
interpret
prove
Formal semantics
Formal proofs

Overview


“If”, “if and only if”, ...
translate
$\rightarrow$, $\leftrightarrow$
interpret
prove
Truth tables for $\rightarrow$
$\rightarrow$Intro, $\rightarrow$Elim

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Let's spend some minutes practicing formal proofs:

$$\neg P \land \neg Q \vdash \neg(P \lor Q)$$

$$\neg(P \lor Q) \vdash \neg P \land \neg Q$$

$$P \lor Q \vdash \neg(\neg P \land \neg Q)$$

(difficult!)

$$\neg(\neg P \land \neg Q) \vdash P \lor Q \qquad \text{and} \qquad \neg(P \land Q) \vdash \neg P \lor \neg Q$$

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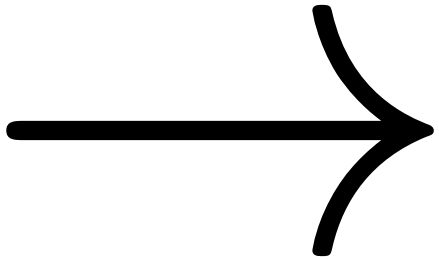
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Material implication in logic

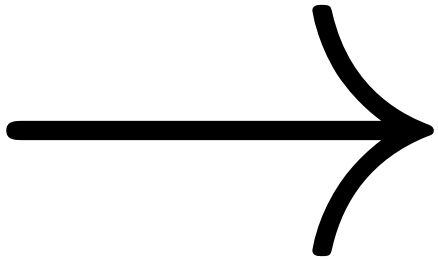


- The sentence $P \to Q$ is a material conditional (or implication);
- We say: "P implies Q"

Truth table:

P	Q	$P \to Q$
T	T	T
T	F	F
F	T	T
F	F	T

Material implication in logic



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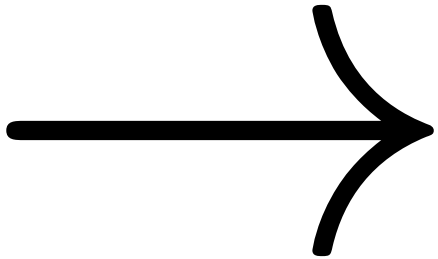
Truth table:

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Compare:

R	S	$R \vee S$
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T	F	T
F	T	T
F	F	F

Material implication in logic



- The sentence $P \to Q$ is a material conditional (or implication);
- We say: "P implies Q"
- Observe: $P \to Q$ has the same truth table as $\neg P \vee Q$. We write $P \to Q \equiv \neg P \vee Q$.

Truth table:

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Compare:

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ordinary reading vs. *logical* reading

Consider the following sentences:

If Groningen is in the Netherlands, then $2 + 2 = 4$.

If the Netherlands is in Europe, then $2 + 2 = 5$.

If Amsterdam is in Belgium, then $2 + 2 = 4$.

If Amsterdam is in Denmark, then $2 + 2 = 5$.

ordinary reading vs. *logical* reading

Consider the following sentences:

If Groningen is in the Netherlands, then $2 + 2 = 4$. $T \to T$

If the Netherlands is in Europe, then $2 + 2 = 5$. $T \to F$

If Amsterdam is in Belgium, then $2 + 2 = 4$. $F \to T$

If Amsterdam is in Denmark, then $2 + 2 = 5$. $F \to F$

Logically OK, but rather unusual in English!

Material Conditional vs Causality

Main points:

- $P \to Q$ carries no causal meaning. It says, if P is true then Q is true.
- $P \to Q$ is true whenever P is false (since $P \to Q \equiv \neg P \vee Q$).

“because” is not truth-functional



Sifan Hassan won gold on the 1500m and 10,000m at the 2019 Athletics World Championships.

- Sifan Hassan won the 1500 meters because she was the fastest.

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  OK, true.
- Sifan Hassan won the 1500 meters because $2 + 2 = 4$.
  ??? not true.

However, *Sifan Hassan was the fastest* and $2 + 2 = 4$ have the same truth value!

So “because” cannot have a truth-functional interpretation: you cannot define “because” with a truth table.

Non truth-functional connectives

- .. because ..
- .. due to ..
- .. as a reaction to ..

Conversational implicature

A meaning of a (natural language) sentence consists of its truth-functional value plus the *conversational implicature*

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Truth-functional: there exists an object called “gas station” and it is located around the corner.
Implied (according to the speakers knowledge): the station is not closed

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Person B can “cancel” the implicature by clarifying: “But I recon the station is closed around this time of day”.

Conversational implicature (ii)

Max is working from home unless Claire is at the library

$$\neg \mathit{Library}(\mathit{claire}) \rightarrow \mathit{Home}(\mathit{max})$$

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But conversational implicature suggests

$$\neg \mathit{Library}(\mathit{claire}) \leftrightarrow \mathit{Home}(\mathit{max})$$

Conversational implicature (iii)

"After the food, you can either have tea or coffee."

Truth-functional meaning: $T \vee C$
Implicature: but not both!

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Truth-functional meaning: $T \vee C$
Implicature: but not both!

".. but of course you can have both drinks if you like"

– we can "cancel" the xor reading afterwards without contradicting ourselves.

Cooperative Principle

Any normal human can understand what we were saying in the previous examples without the use of logic. So how does it work?

Answer by Grice (1975): in a conversation we follow the cooperative principle

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Answer by Grice (1975): in a conversation we follow the cooperative principle

Conversation between two (rational) agents, cooperating to achieve a common goal (or each their own goals).
The principle is split into several maxims.

Maxim of Quality

Believe what you say.

Or, better, do not say what you believe to be false.

Maxim of Quantity

Give the most optimal and helpful amount of information.

No more and no less!

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Give the most optimal and helpful amount of information.

No more and no less!

Person A (looking for the restroom): excuse me, do you know where the toilet is?
Person B: yes, I do know

Maxim of Relation

Say things that are relevant.

Or, do not say things that are not relevant.

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Say things that are relevant.

Or, do not say things that are not relevant.

Person A: Will you come to my party tomorrow?
Person B: I have an early exam the day after.

Maxim of Manner

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- "She got pregnant and had a child" vs "She had a child and got pregnant"

Maxim of Manner

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This maxim is especially useful for writing, but also helpful in speech.

- "I went to a concert. The musicians produced various sounds resembling the melody of Stairway to Heaven".
- "She got pregnant and had a child" vs "She had a child and got pregnant"
- "Today I am going to talk about conversational implicature. It is a way of understanding meaning of sentences beyond truth-value semantics."

More examples

- “He stopped the car” vs “he caused the car to stop”

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- A: “Did you finish your homework already?”
  B: “Well, Eldon Ring Nightreign came out last week”

More examples

- "He stopped the car" vs "he caused the car to stop"

- A: "Will you come to my party?"
  B: "It is not impossible for me to come"

- A: "Did you finish your homework already?"
  B: "Well, Eldon Ring Nightreign came out last week"
- A: "Did you spend all of your salary on Funko Pops?"
  B: "I spend a part of my salary on that"

Flouting and violating the maxims

Violating the maxims means that we are (sub)consciously breaking these principles, without cooperation of other people.
Violating the maxim of quality is lying, violating the maxim of quantity is lying by omission.

Flouting is when you break the maxims in a wink-wink nudge-nudge way, that is more or less obvious to everyone. Can denote humor, irony, or just emphasis and metaphors.

Flouting Maxims for Effects

When we flout a maxim we break it in such an obvious way that the interlocutor cannot take it at a face value and forces to infer more information.

- "I went to a concert. The musicians produced various sounds resembling the melody of Stairway to Heaven"

- A: How did the lecture go?
  B: The coffee during the break was very nice.

Flouting Maxims for Giggles

- “One morning I shot an elephant in my pajamas”
- “How did an elephant got into your pajamas?”

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Two people in the park:

- A: “Does you dog bite?”
- B: “No, of course not.”

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- B: “No, of course not.”
- When person A tries to pet the dog, the dog nearly bites his hand off.
- A: “I thought you said your dog doesn’t bite”, said the old man, blood dripping from his hand.
- B: “That’s right. My dog doesn’t bite, but that’s not my dog.”

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Fitch rules for conditionals

Question: Which new Fitch rules are we going to see in the upcoming slides?

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Answer: $\rightarrow$ Intro and $\rightarrow$ Elim

Conditional Proof (example)

Claim: If n is even, then $3n$ is even.

Proof.

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- Suppose that n is even.

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- It follows that $3n = 3 \cdot 2k = 2 \cdot 3k$, so $3n$ is even, too.

Conditional Proof (example)

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Proof.

- Suppose that n is even.
- That means n is a multiple of 2, i.e. $n = 2k$ for some k.
- It follows that $3n = 3 \cdot 2k = 2 \cdot 3k$, so $3n$ is even, too.

So the claim has been proved: if n is even, then $3n$ is even.

Conditional proof (general form)

General form of a conditional proof:

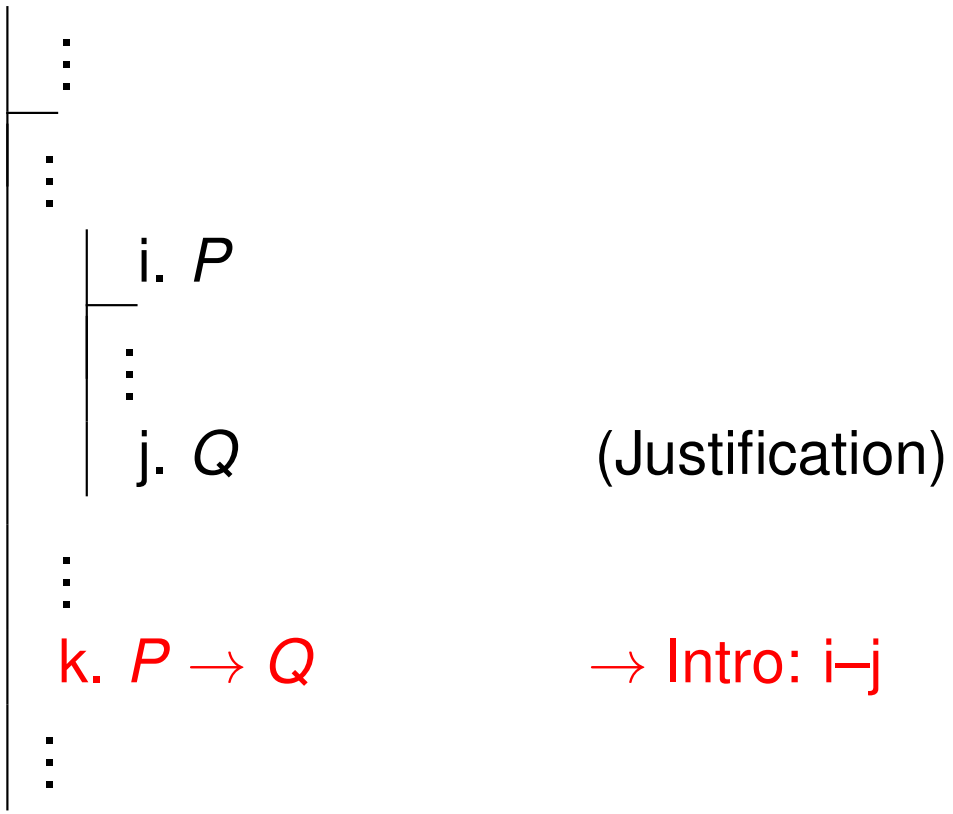
Suppose that P.

. . . [some reasoning] . . .

Then Q.

Conclusion: if P then Q.

→ Introduction



Conditional proof:
Assuming P, prove Q.

Modus Ponens

If you are in Groningen, then you are in the Netherlands.
You are in Groningen.

You are in the Netherlands.

This is an instance of Modus Ponens ($\rightarrow$ Elimination):

$P \rightarrow Q$
P

Q

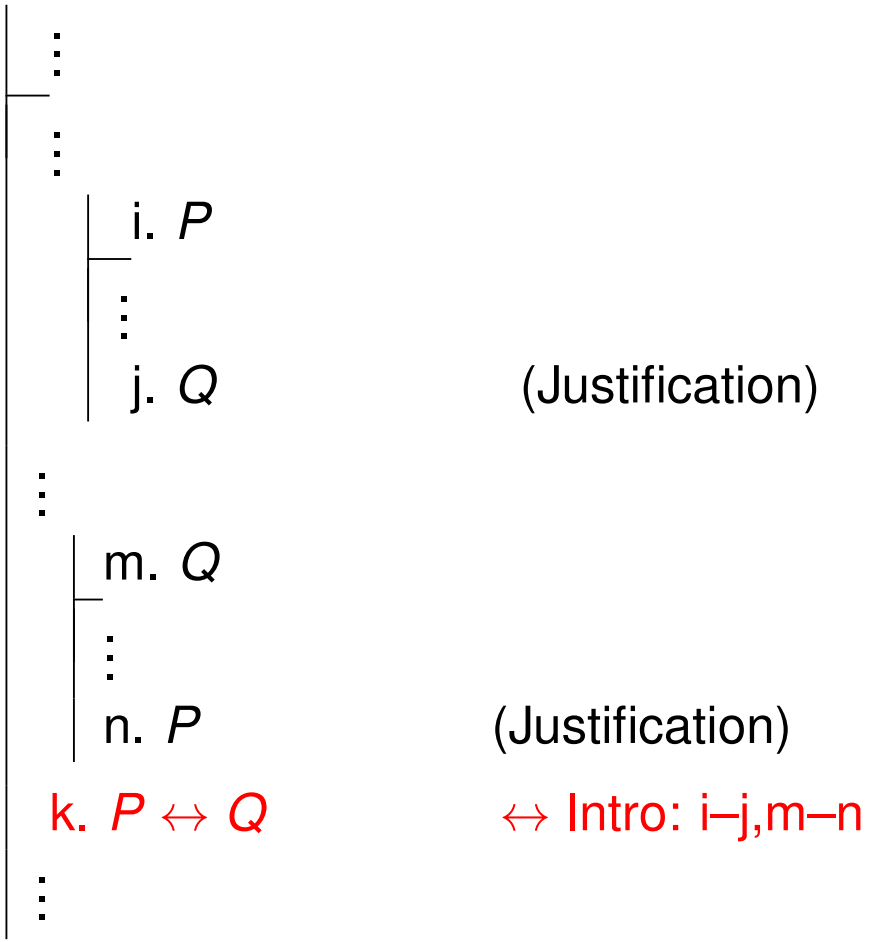
$\rightarrow$ Elimination (Modus Ponens)

"From $P \rightarrow Q$ and P, we infer Q".

⋮		
i.	$P \rightarrow Q$	Justification (or premise)
⋮		
j.	P	Justification (or premise)
⋮		
k.	Q	$\rightarrow$ Elim: i,j
⋮		

Note the order of references in the justification: first $P \rightarrow Q$, then P.

$\leftrightarrow$ Introduction



Recall that:
$P \leftrightarrow Q \equiv (P \rightarrow Q) \wedge (Q \rightarrow P)$.

To prove $P \leftrightarrow Q$,
construct subproofs for
$P \rightarrow Q$ and $Q \rightarrow P$.

Note the order of references in the justification is first $P \rightarrow Q$ and then $Q \rightarrow P$.

$\leftrightarrow$ Elimination

"From $P \leftrightarrow Q$ and P, we infer Q"

⋮		
i.	$P \leftrightarrow Q$	Justific. (or Premise)
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⋮		
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⋮		

Again: Note the order of references in the justification is first $P \leftrightarrow Q$ and then P (or Q).

Example 1: Transitivity

If you are in Groningen, then you are in the Netherlands.
If you are in the Netherlands, then you are on Planet Earth.

If you are in Groningen, then you are on Planet Earth.

How would you formalize and prove it in PL?

Example 1: proving $P \to Q, Q \to R \vdash P \to R$

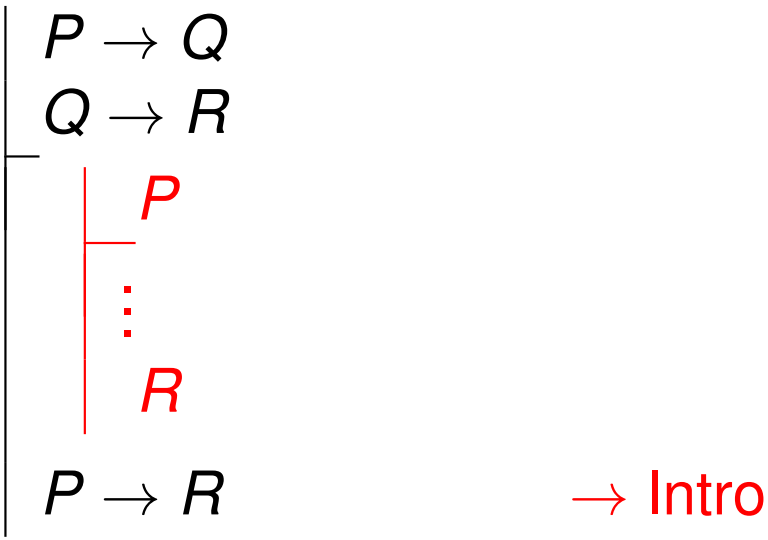
$$\begin{array}{l} P \to Q \\ Q \to R \\ \hline \vdots \\ P \to R \end{array}$$

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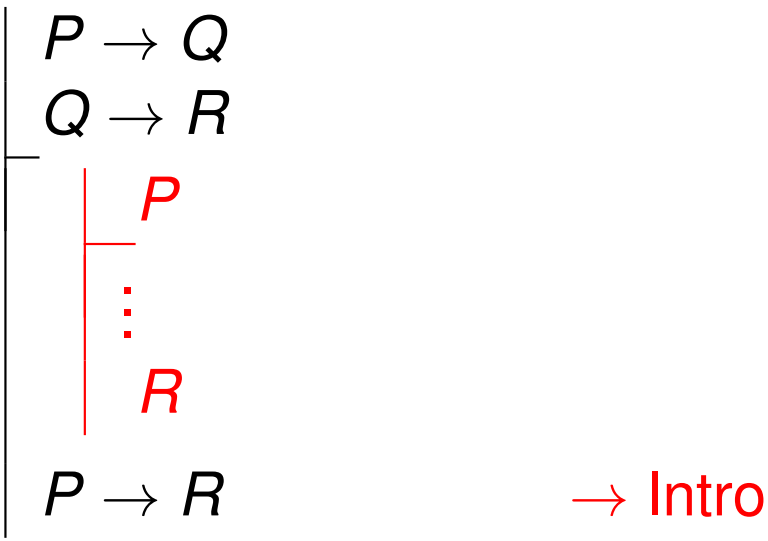
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Idea: use $\to$ Introduction

Example 1: proving $P \to Q, Q \to R \vdash P \to R$



Example 1: proving $P \to Q, Q \to R \vdash P \to R$



Idea: use $\to$ Elimination twice

Example 1: proving $P \to Q, Q \to R \vdash P \to R$

$P \to Q$
$Q \to R$
$\quad P$
$\quad Q$ $\to$ Elim
$\quad R$ $\to$ Elim
$P \to R$ $\to$ Intro

Example 1: proving $P \to Q, Q \to R \vdash P \to R$

1.	$P \to Q$	
2.	$Q \to R$	
3.	P	
4.	Q	$\to$ Elim: 1, 3
5.	R	$\to$ Elim: 2, 4
6.	$P \to R$	$\to$ Intro: 3–5

Example 2: from disjunction to implication

You are either not in Groningen, or you are in the Netherlands

If you are in Groningen, then you are in the Netherlands

How to formalize this in PL?

Example 2: proving $\neg P \vee Q \vdash P \rightarrow Q$

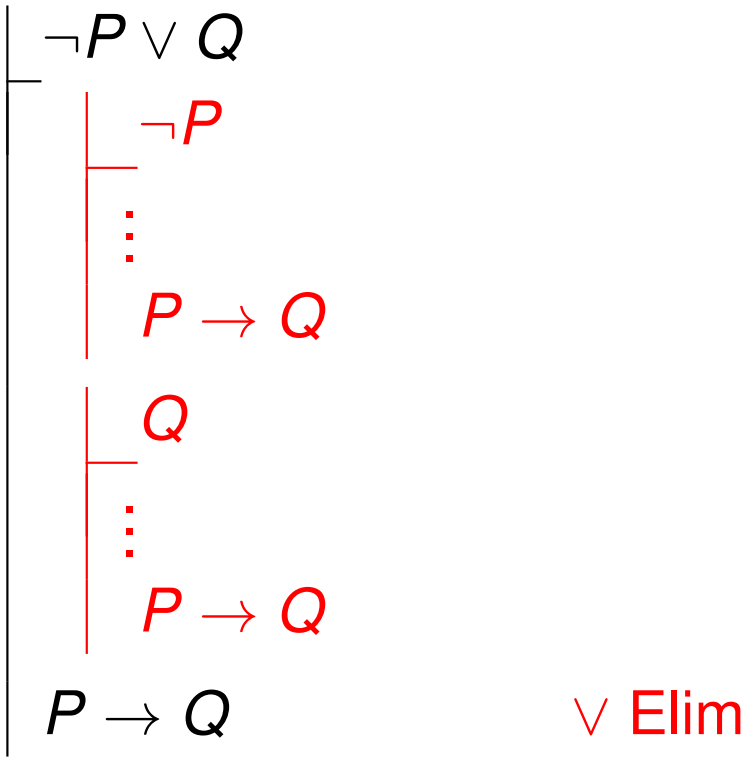
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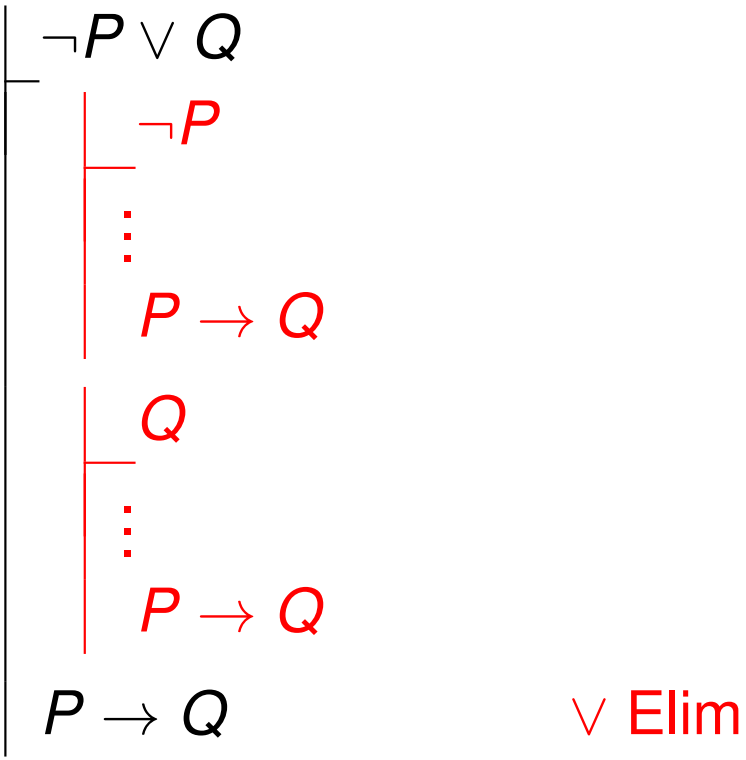
$$\begin{array}{|l} \neg P \vee Q \\ \hline \vdots \\ P \rightarrow Q \end{array}$$

Idea: use $\vee$ Elimination

Example 2: proving $\neg P \vee Q \vdash P \rightarrow Q$

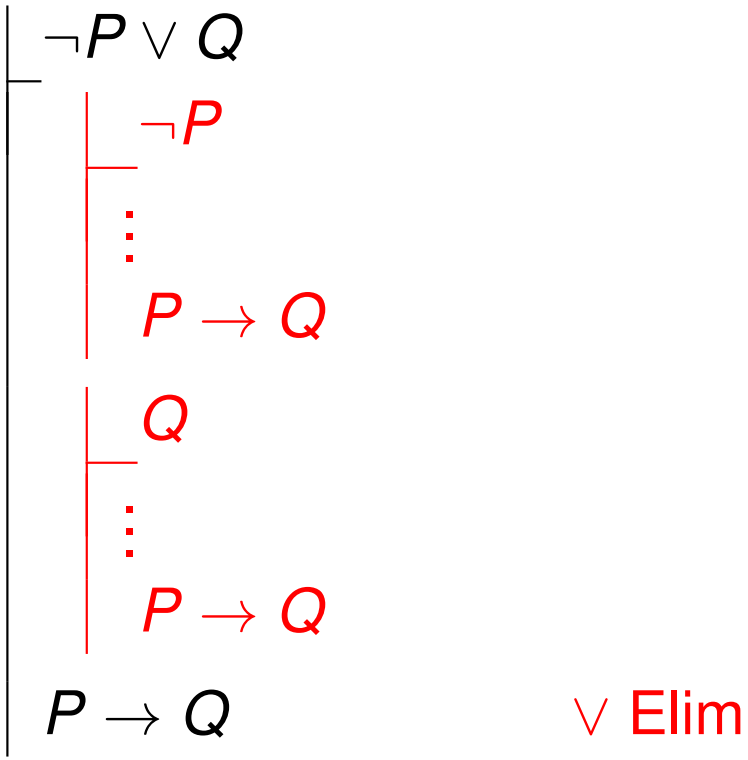


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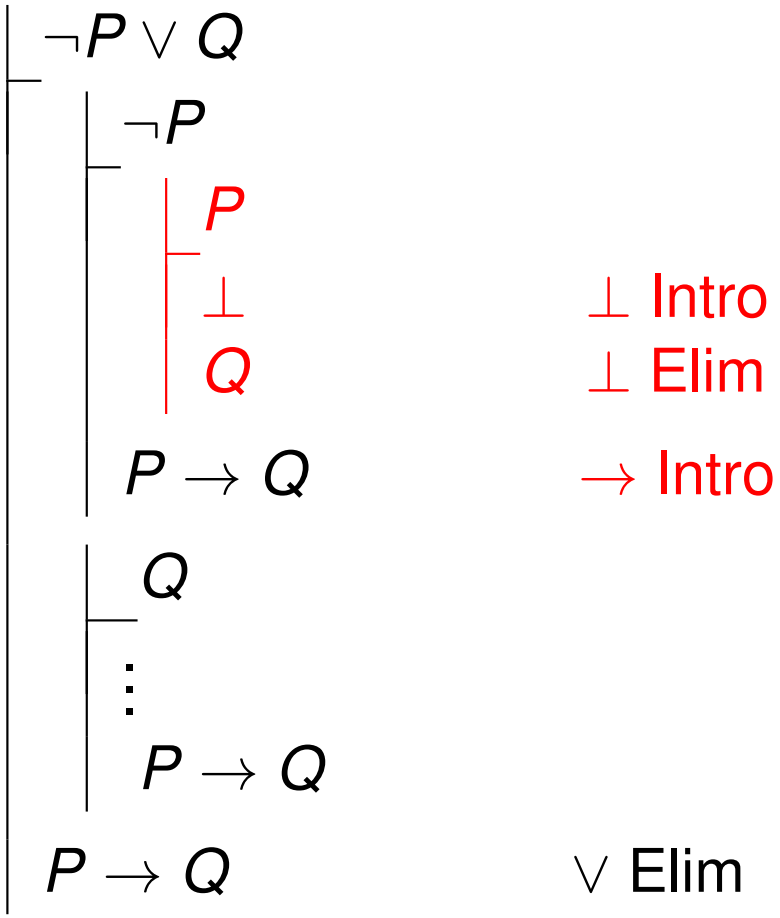
Idea:
To prove $P \rightarrow Q$ follows from $\neg P$:

Example 2: proving $\neg P \vee Q \vdash P \rightarrow Q$

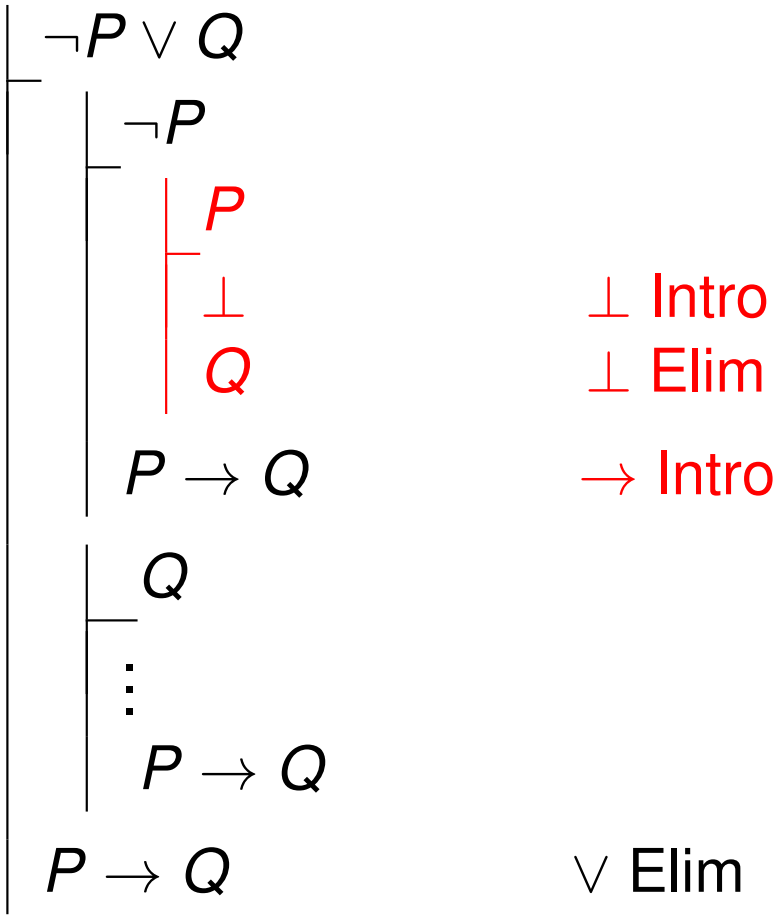


Idea:
To prove $P \rightarrow Q$ follows from $\neg P$:
If we assume P, then P and $\neg P$ lead to $\bot$,
from which everything follows, including Q.

Example 2: proving $\neg P \vee Q \vdash P \rightarrow Q$

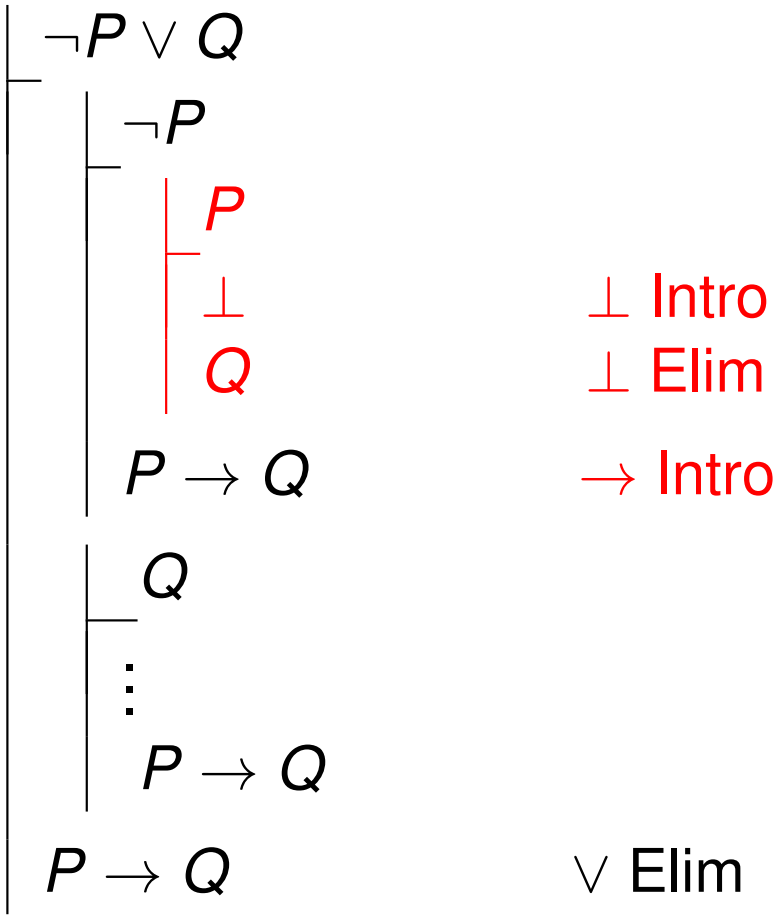


Example 2: proving $\neg P \vee Q \vdash P \to Q$



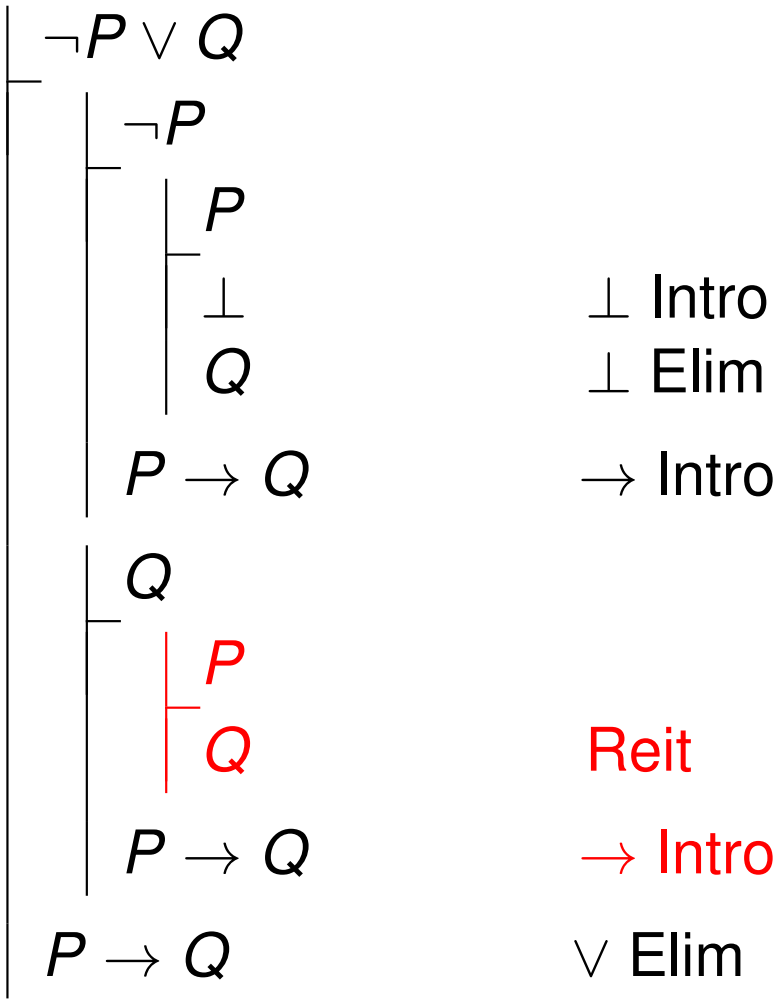
Idea:
To prove $P \to Q$ follows from Q:

Example 2: proving $\neg P \vee Q \vdash P \to Q$



Idea:
To prove $P \to Q$ follows from Q:
obtain Q by Reiteration, even
without using assumption P

Example 2: proving $\neg P \vee Q \vdash P \rightarrow Q$



Example 2: proving $\neg P \vee Q \vdash P \rightarrow Q$

1. $\neg P \vee Q$	
2. $\neg P$	
3. P	
4. $\bot$	$\bot$ Intro: 3, 2
5. Q	$\bot$ Elim: 4
6. $P \rightarrow Q$	$\rightarrow$ Intro: 3–5
7. Q	
8. P	
9. Q	Reit: 7
10. $P \rightarrow Q$	$\rightarrow$ Intro 8–9
11. $P \rightarrow Q$	$\vee$ Elim: 1, 2–6, 7–10

Example 3: proving $P \to Q \vdash \neg P \vee Q$

$P \to Q$

$\vdots$

$\neg P \vee Q$

Example 3: proving $P \to Q \vdash \neg P \vee Q$

$$\begin{array}{|l} P \to Q \\ \hline \quad \vdots \\ \quad \neg P \vee Q \end{array}$$

Idea: try to obtain $\neg\neg(\neg P \vee Q)$ and apply $\neg$ Elimination.

Example 3: proving $P \to Q \vdash \neg P \vee Q$

$P \to Q$

$\vdots$

$\neg\neg(\neg P \vee Q)$

$\neg P \vee Q$ $\neg$ Elim

Example 3: proving $P \to Q \vdash \neg P \vee Q$

$P \to Q$

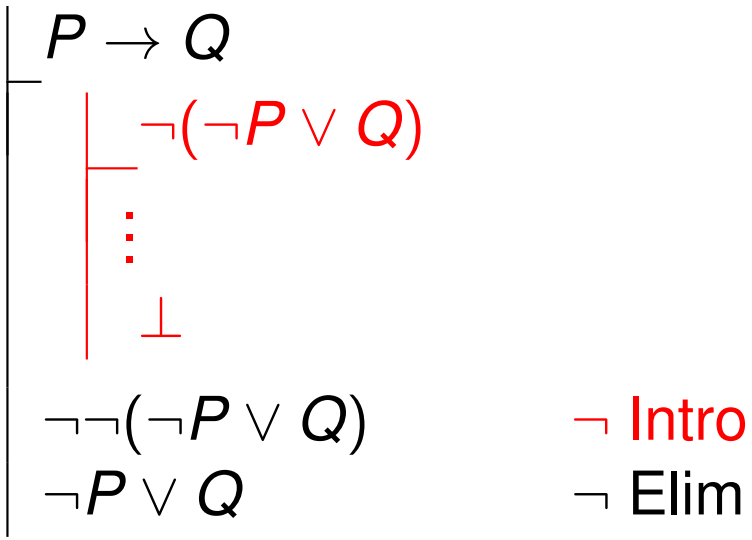
$\vdots$

$\neg\neg(\neg P \vee Q)$

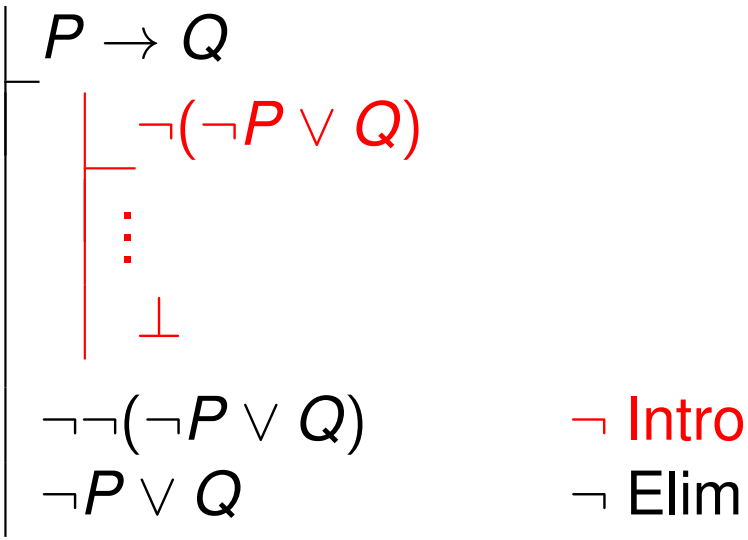
$\neg P \vee Q$ $\neg$ Elim

Idea: use $\neg$ Intro to obtain $\neg\neg(\neg P \vee Q)$.

Example 3: proving $P \to Q \vdash \neg P \vee Q$



Example 3: proving $P \to Q \vdash \neg P \vee Q$



Idea: try to obtain $\neg P \vee Q$, then we have $\bot$ via $\bot$ Introduction.

Example 3: proving $P \to Q \vdash \neg P \vee Q$

$P \to Q$

$\neg(\neg P \vee Q)$	
$\vdots$	
$\neg P \vee Q$	
$\bot$	$\bot$ Intro

$\neg\neg(\neg P \vee Q)$ $\neg$ Intro

$\neg P \vee Q$ $\neg$ Elim

Example 3: proving $P \to Q \vdash \neg P \vee Q$

$P \to Q$

$\neg(\neg P \vee Q)$

$\vdots$

$\neg P \vee Q$

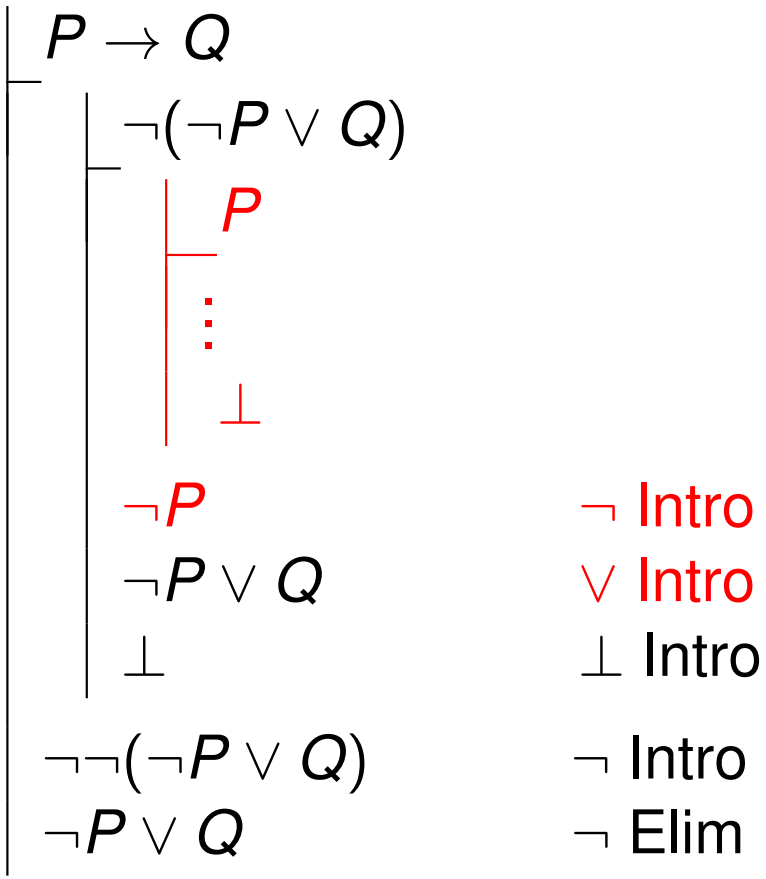
$\bot$ $\bot$ Intro

$\neg\neg(\neg P \vee Q)$ $\neg$ Intro

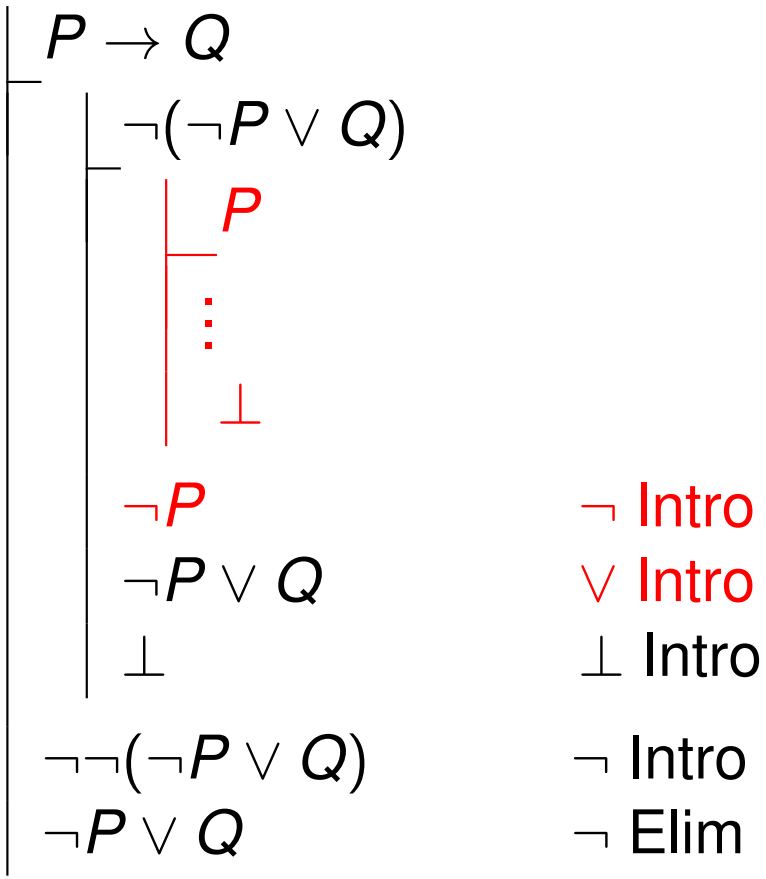
$\neg P \vee Q$ $\neg$ Elim

Idea: try to obtain $\neg P$, then we have $\neg P \vee Q$ via $\vee$ Introduction.
By $\neg$ Introduction, it suffices to prove $\bot$ from the assumption P.

Example 3: proving $P \to Q \vdash \neg P \vee Q$

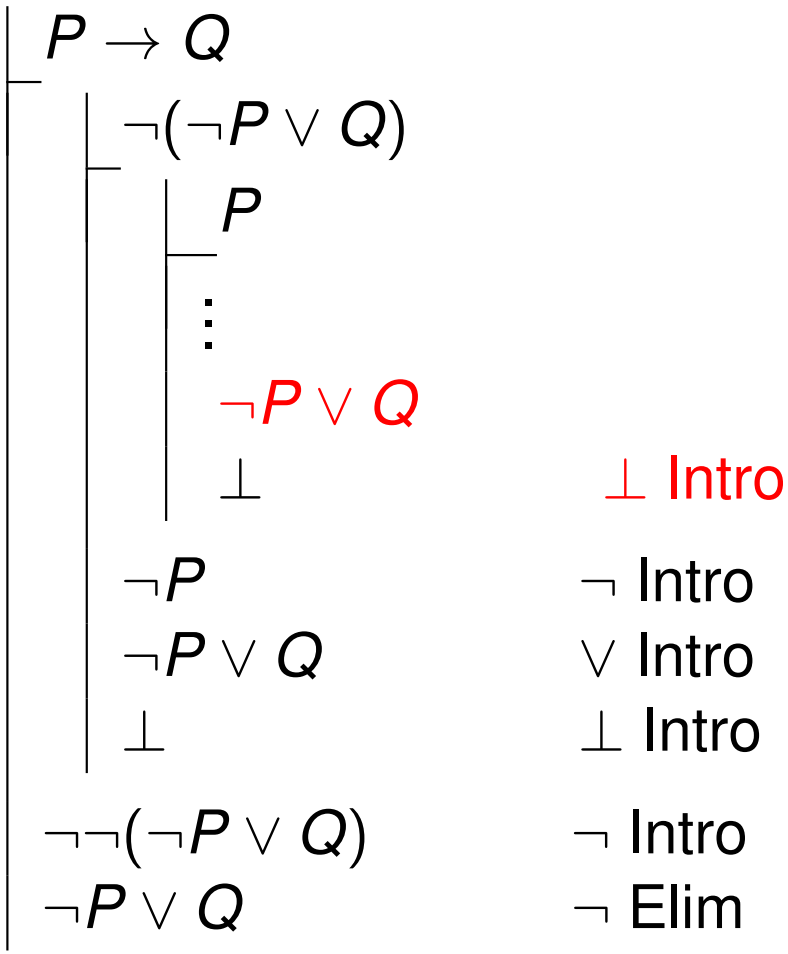


Example 3: proving $P \to Q \vdash \neg P \vee Q$

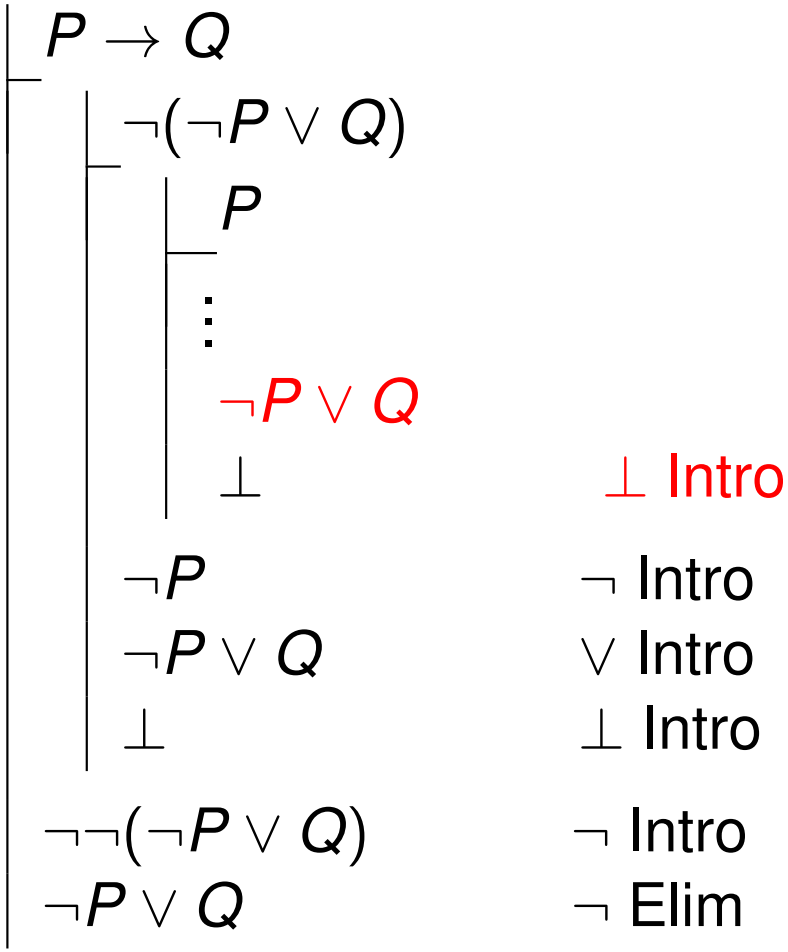


Idea: try to obtain $\neg P \vee Q$,
then we have $\bot$ via $\bot$ Introduction
and $\neg(\neg P \vee Q)$.

Example 3: proving $P \to Q \vdash \neg P \vee Q$

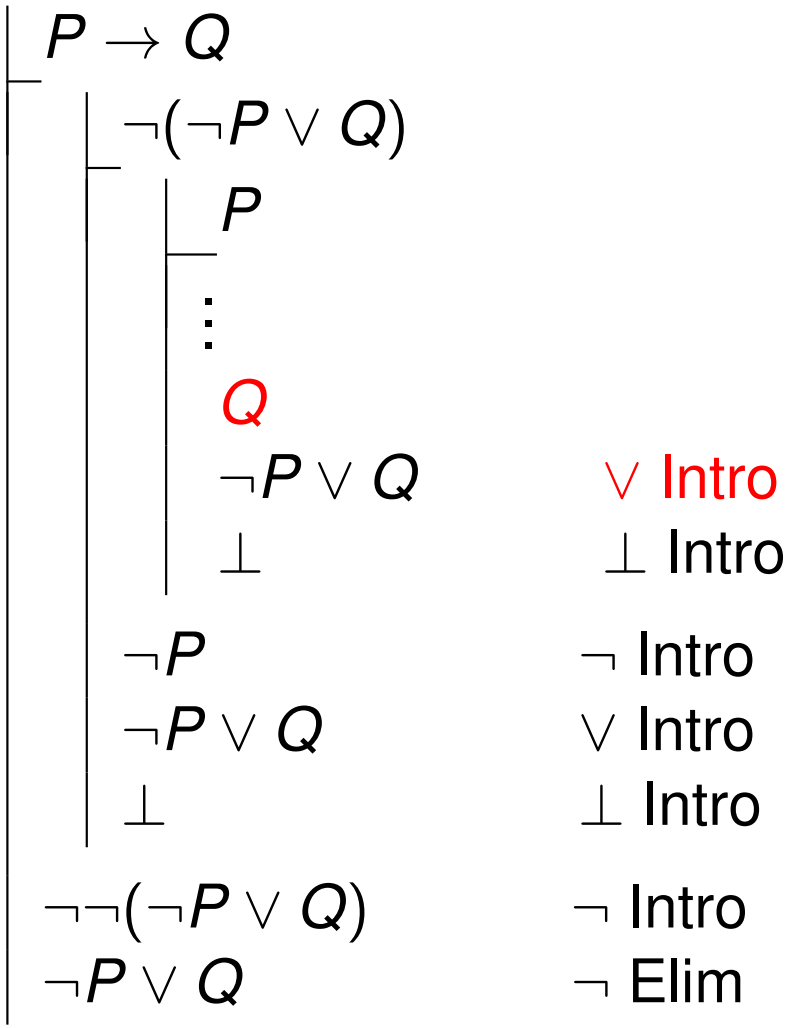


Example 3: proving $P \to Q \vdash \neg P \vee Q$

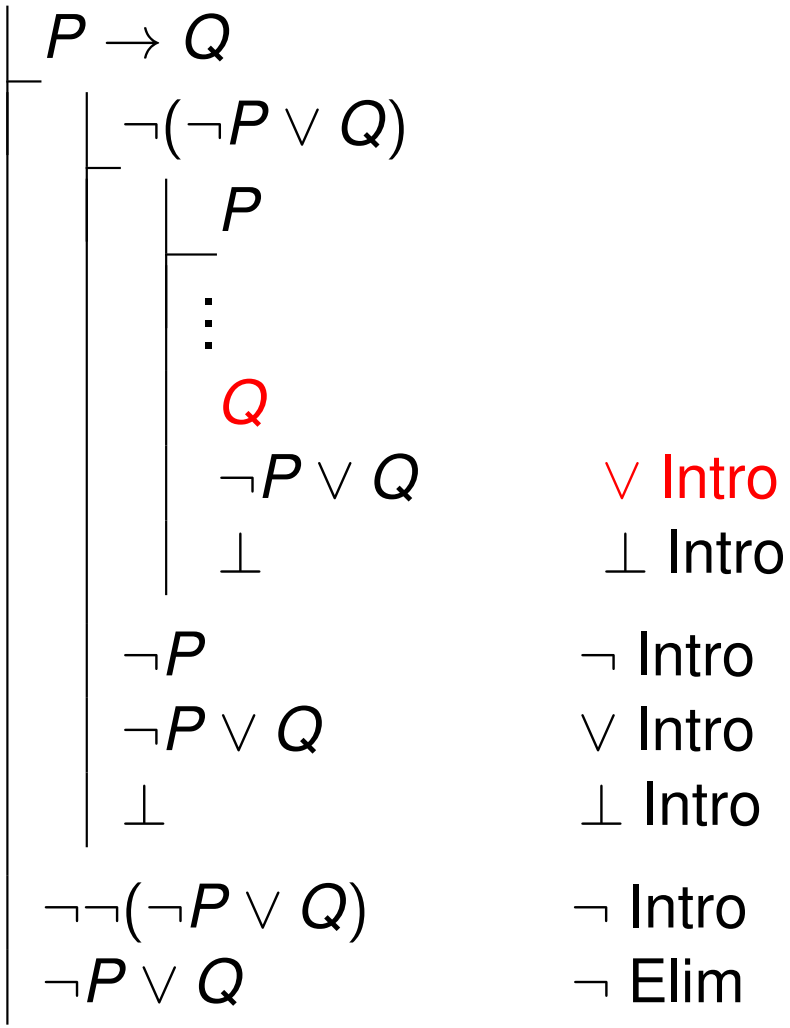


How to obtain $\neg P \vee Q$?
From Q by $\vee$ Introduction.

Example 3: proving $P \to Q \vdash \neg P \vee Q$



Example 3: proving $P \to Q \vdash \neg P \vee Q$



How to obtain Q?
From the assumptions P and $P \to Q$, using $\to$ Elimination.

Example 3: proving $P \to Q \vdash \neg P \vee Q$

$P \to Q$	
$\quad \neg(\neg P \vee Q)$	
$\quad\quad P$	
$\quad\quad Q$	$\to$ Elim
$\quad\quad \neg P \vee Q$	$\vee$ Intro
$\quad\quad \bot$	$\bot$ Intro
$\quad \neg P$	$\neg$ Intro
$\quad \neg P \vee Q$	$\vee$ Intro
$\quad \bot$	$\bot$ Intro
$\neg\neg(\neg P \vee Q)$	$\neg$ Intro
$\neg P \vee Q$	$\neg$ Elim

Example 3: proving $P \to Q \vdash \neg P \vee Q$

1.	$P \to Q$	
2.	$\neg(\neg P \vee Q)$	
3.	P	
4.	Q	$\to$ Elim: 1, 3
5.	$\neg P \vee Q$	$\vee$ Intro: 4
6.	$\bot$	$\bot$ Intro: 5, 2
7.	$\neg P$	$\neg$ Intro: 3–6
8.	$\neg P \vee Q$	$\vee$ Intro: 7
9.	$\bot$	$\bot$ Intro: 8, 2
10.	$\neg\neg(\neg P \vee Q)$	$\neg$ Intro: 2–9
11.	$\neg P \vee Q$	$\neg$ Elim: 10

Alternative proof of $P \to Q \vdash \neg P \vee Q$

1.	$P \to Q$	
2.	$\neg(\neg P \vee Q)$	
3.	$\neg P$	
4.	$\neg P \vee Q$	$\vee$ Intro: 3
5.	$\bot$	$\bot$ Intro: 4, 2
6.	$\neg\neg P$	$\neg$ Intro: 3–5
7.	P	$\neg$ Elim: 6
8.	Q	$\to$ Elim: 1, 7
9.	$\neg P \vee Q$	$\vee$ Intro: 8
10.	$\bot$	$\bot$ Intro: 9, 2
11.	$\neg\neg(\neg P \vee Q)$	$\neg$ Intro 2–10
12.	$\neg P \vee Q$	$\neg$ Elim: 11

Today's Lecture

1 Overview

2 Proof practice

3 Conversational implicature

4 Proofs with (bi)conditionals

5 Truth tables, tautologies and logical truths

6 Concluding

Truth Tables for Complex Formulas

- Let P be a complex formula P constructed from atomic propositions $A_1, \ldots, A_n$ and Boolean connectives (and conditionals).

  Example: $\neg(A \wedge (\neg A \vee (B \wedge C))) \vee B$

- How do we evaluate it?

Truth Tables for Complex Formulas

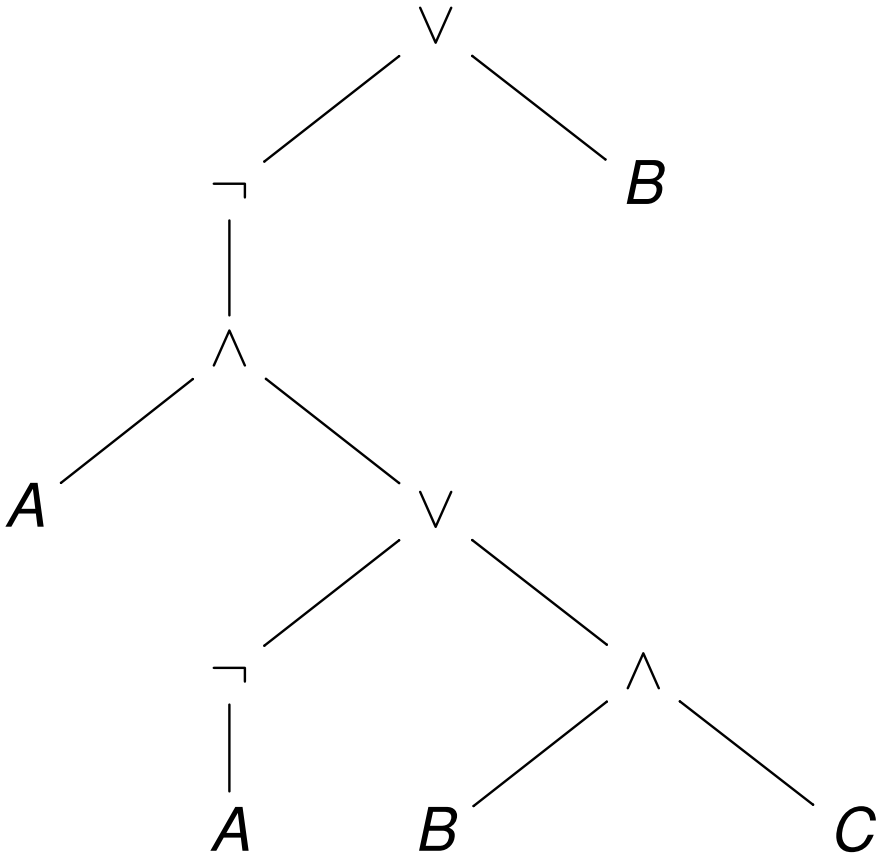
- Let P be a complex formula P constructed from atomic propositions $A_1, \ldots, A_n$ and Boolean connectives (and conditionals).

  Example: $\neg(A \wedge (\neg A \vee (B \wedge C))) \vee B$

- How do we evaluate it?
- The truth table of P is obtained by applying the truth tables of the connectives "bottom-up".
- Sometimes its useful to visualize the *syntax tree* of a complex formula

Syntax Tree by Example

Syntax tree of $\neg(A \wedge (\neg A \vee (B \wedge C))) \vee B$



making a truth table

A	B	C	$\neg(A \wedge (\neg A \vee (B \wedge C))) \vee B$
T	T	T	
T	T	F	
T	F	T	
T	F	F	
F	T	T	
F	T	F	
F	F	T	
F	F	F	

making a truth table

A	B	C	$\neg(A \wedge (\neg A \vee (B \wedge C))) \vee B$
T	T	T	F
T	T	F	F
T	F	T	F
T	F	F	F
F	T	T	T
F	T	F	T
F	F	T	T
F	F	F	T

making a truth table

A	B	C	$\neg(A \wedge (\neg A \vee (B \wedge C))) \vee B$
T	T	T	F T
T	T	F	F F
T	F	T	F F
T	F	F	F F
F	T	T	T T
F	T	F	T F
F	F	T	T F
F	F	F	T F

making a truth table

A	B	C	$\neg(A \wedge (\neg A \vee (B \wedge C))) \vee B$
T	T	T	F T T
T	T	F	F F F
T	F	T	F F F
T	F	F	F F F
F	T	T	T T T
F	T	F	T T F
F	F	T	T T F
F	F	F	T T F

making a truth table

A	B	C	$\neg(A \wedge (\neg A \vee (B \wedge C))) \vee B$
T	T	T	T F T T T
T	T	F	F F F F F
T	F	T	F F F F F
T	F	F	F F F F F
F	T	T	F T T T T
F	T	F	F T T F F
F	F	T	F T T F F
F	F	F	F T T F F

making a truth table

A	B	C	$\neg(A \wedge (\neg A \vee (B \wedge C))) \vee B$
T	T	T	F T F T T
T	T	F	T F F F F
T	F	T	T F F F F
T	F	F	T F F F F
F	T	T	T F T T T
F	T	F	T F T T F
F	F	T	T F T T F
F	F	F	T F T T F

making a truth table

A	B	C	$\neg(A$	$\wedge$	$(\neg A$	$\vee$	$(B \wedge C)))$	$\vee$	B
T	T	T	F	T	F	T	T	T	
T	T	F	T	F	F	F	F	T	
T	F	T	T	F	F	F	F	T	
T	F	F	T	F	F	F	F	T	
F	T	T	T	F	T	T	T	T	
F	T	F	T	F	T	T	F	T	
F	F	T	T	F	T	T	F	T	
F	F	F	T	F	T	T	F	T	

making a truth table

A	B	C	$\neg(A$	$\wedge$	$(\neg A$	$\vee$	$(B \wedge C)))$	$\vee$	B
T	T	T	F	T	F	T	T	T	
T	T	F	T	F	F	F	F	T	
T	F	T	T	F	F	F	F	T	
T	F	F	T	F	F	F	F	T	
F	T	T	T	F	T	T	T	T	
F	T	F	T	F	T	T	F	T	
F	F	T	T	F	T	T	F	T	
F	F	F	T	F	T	T	F	T	
(1)	(2)	(3)	(8)	(7)	(4)	(6)	(5)	(9)	

making a truth table

A	B	C	$\neg(A$	$\wedge$	$(\neg A$	$\vee$	$(B \wedge C)))$	$\vee$	B
T	T	T	F	T	F	T	T	T	
T	T	F	T	F	F	F	F	T	
T	F	T	T	F	F	F	F	T	
T	F	F	T	F	F	F	F	T	
F	T	T	T	F	T	T	T	T	
F	T	F	T	F	T	T	F	T	
F	F	T	T	F	T	T	F	T	
F	F	F	T	F	T	T	F	T	
(1)	(2)	(3)	(8)	(7)	(4)	(6)	(5)	(9)	

Observe that $\neg(A \wedge (\neg A \vee (B \wedge C))) \vee B$ always has truth value $\top$.
We call such a sentence a **tautology**.

Tautologies and Logical Truths

Definition:

- A sentence *S* is a tautology if in the truth table for *S*, there are only T's in the column under the main connective of *S*.
  (*S* is true for all possible truth values of its atomic parts.)

- A sentence *S* is a logical truth (or: logical necessity) if it true under all logically possible circumstances.

So **every tautology is a logical truth**.

Tautologies and Logical Truths

But: a logical truth is not always a tautology.

Consider $a = a \vee a = b$. It has the truth table

$a = a$	$a = b$	$a = a \vee a = b$
T	T	T
T	F	T
F	T	T
F	F	F

so it is not a tautology.

However, it is a logical truth.

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T	T	T
T	F	T
F	T	T
F	F	F

so it is not a tautology.

However, it is a logical truth.
The last two rows refer to situations that are *logically impossible*.

Spurious Rows

Definition: A row in a truth table is called a spurious row if that row corresponds to a logically impossible situation due to the meaning of the atomic sentences in the formula.

We have: A sentence S is a logical truth if in the truth table of S there are only T's in all non-spurious rows in the main connective column.

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We have: A sentence S is a logical truth if in the truth table of S there are only T's in all non-spurious rows in the main connective column.

In Propositional Logic without equality, there are no spurious rows, and the notions of *logical truth* and *tautology* coincide.

Contradictions

Definition:

- A sentence S is a tautology if in the truth table for S, there are only T's in the main connective column.
  (S is true for all possible truth values of its atomic parts.)
- A sentence S is a logical truth (or: logical necessity) if in the truth table for S, there are only T's in all non-spurious rows in the main connective column.
- A sentence S is a (TT-)contradiction if
- A sentence S is a

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Contradictions

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- A sentence S is a (TT-)contradiction if in the truth table for S, there are only F's in the main connective column.
- A sentence S is a logical contradiction if in the truth table for S, there are only F's in all non-spurious rows in the main connective column.

Example

Consider $a \neq a$ (shorthand for $\neg(a = a)$)

$a = a$	$\neg(a = a)$	
T	F	
F	T	(*spurious*)

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$a = a$	$\neg(a = a)$	
T	F	
F	T	(*spurious*)

$a \neq a$ is a logical contradiction, but not a TT-contradiction

Quick question

Which of the following are what?

- $P \vee \neg P$
- $a = b \rightarrow (b = c \rightarrow a \neq c)$
- $P \wedge \neg P$
- $P \wedge Q$

TT-possible

Definition. A sentence *S* is called TT-possible if in the truth table for *S*, there is at least one T in the column under the main connective of *S*.

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Note that for each sentence S:

- S is a tautology iff $\neg S$ is a contradiction;

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Note that for each sentence S:

- S is a tautology iff $\neg S$ is a contradiction;
- S is TT-possible iff S is not a contradiction;
- S is a tautology iff $\neg S$ is not TT-possible

Tautologies, Contradictions and Possibilities

Sentence is ...	if main connective column in truth table has...
a *tautology*	T in all rows
a *logical truth*	T in all non-spurious rows
a *(TT-)contradiction*	F in all rows
a *logical contradiction*	F in all non-spurious rows
TT-possible	T in at least one row
logically possible	T in at least one non-spurious row

Logical vs Tautological in PL and FOL

In PL (without =, where atomic sentences as letters P, Q, ...),

- tautologies and logical truths coincide.
- Similarly, for contradiction and possibility notions.

In FOL (with = and relation symbols):

- All tautologies are logical truths, but not vice versa.
- All TT-contradictions are logical contradictions, but not vice versa.
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- All logically possible sentences are TT-possible, but not vice versa.

By looking at those definitions from a different aspect, we can understand the differences more sharply.

The Return of Fitch

"Horizontal" Fitch notation: $P_1, \ldots, P_n \vdash Q$ should be read as

P_1
$\vdots$
P_n
Q

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We don't have a notion of tautology/contradiction, but we have a notion of validity.

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P_1
$\vdots$
P_n

Q

We don't have a notion of tautology/contradiction, but we have a notion of validity.

Recall (definition of of validity): The following are equivalent:

- The argument $P_1, \ldots, P_n \vdash Q$ is *valid*
- Q is a logical consequence of $P_1, \ldots, P_n$
- In all possible situations in which all $P_1, \ldots, P_n$ are true, also Q is true.

Logical consequence and Logical necessity

Fact 1: the following are equivalent:

1. Q is a logical consequence of $P_1, \ldots, P_n$;
2. There is a proof in system $\mathcal{F}$ of $P_1, \ldots, P_n \vdash Q$.

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How to use this? To establish that $P_1, \ldots, P_n \vdash Q$ is valid, check a truth table condition (item 3), or give a formal proof (item 2).

Note that:

- item 3 is a semantic notion (based on truth tables)
- items 1-2 is a syntactic, proof-theoretic notion

Tautological consequence

Definition: Q is a tautological consequence of $P_1, \ldots, P_n$ (write: $P_1, \ldots, P_n \Vdash Q$) if
in every row in the truth table where $P_1, \ldots, P_n$ are all true, Q is also true.

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Fact 2: The following are equivalent:

- Q is a tautological consequence of $P_1, \ldots, P_n$;
- the sentence $(P_1 \wedge \cdots \wedge P_n) \to Q$ is a tautology.

For PL sentences, tautological consequence coincides with logical consequence. This does not hold for FOL sentences.
We have

- $a = b \vdash b = a$,
- but $b = a$ is not a tautological consequence of $a = b$.

Logical and Tautological Equivalence

(Tauto)logical consequence goes one way: from premise to conclusion. There are bi-directional versions:

Definition:

- P and Q are logically equivalent (notation: $P \Leftrightarrow Q$) if $P \vdash Q$ and $Q \vdash P$ are both valid.
- P and Q are tautologically equivalent (notation: $P \equiv Q$) if in every row of the joint truth table, P and Q receive the same truth value.

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Fact 3:

- $P \Leftrightarrow Q$ iff the sentence $P \leftrightarrow Q$ is a logical truth.
- $P \equiv Q$ iff the sentence $P \leftrightarrow Q$ is a tautology.
- If P and Q are PL sentences, then: $P \iff Q$ iff $P \equiv Q$.

Examples of logical equivalences

Logical equivalence is one of the "strongest" notions of equivalence between two propositions; we will return to it in the second half of the lecture

law of double negation	P	$\Leftrightarrow$	$\neg\neg P$
definability of $\rightarrow$	$P \rightarrow Q$	$\Leftrightarrow$	$\neg P \vee Q$
definability of $\leftrightarrow$	$P \leftrightarrow Q$	$\Leftrightarrow$	$(P \rightarrow Q) \wedge (Q \rightarrow P)$

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definability of $\leftrightarrow$	$P \leftrightarrow Q$	$\Leftrightarrow$	$(P \rightarrow Q) \wedge (Q \rightarrow P)$

Meta-level vs object-level

Important:

- The symbols $\Leftrightarrow$, $\equiv$ is meta-notation; it is not a relation symbol in FOL/PL. In particular, $P \leftrightarrow Q$ is not a sentence in FOL/PL. (Meta notation vs object notation.)
- Logical equivalences cannot be used as proof steps in system $\mathcal{F}$ proofs. Formal proofs are syntactic objects. Different formulas require different rules. (The Fitch tool from the book allows you to use them via a special rule)

Summarizing the notions

Single proposition	Multiple propositions	
Tautology	Taut. Consequence $P \Vdash Q$	Taut. Equivalence $P \equiv Q$
Logical Truth	Log. Consequence $P \vdash Q$	Log. Equivalence $P \Leftrightarrow Q$

Summarizing the notions

Single proposition	Multiple propositions	
Tautology	Taut. Consequence $P \Vdash Q$	Taut. Equivalence $P \equiv Q$
Logical Truth	Log. Consequence $P \vdash Q$	Log. Equivalence $P \Leftrightarrow Q$

No need to memorize this like a poem, just remember the axis along which the notions change.

Checking consequences, equivalence

Suppose you are asked to check whether two sentences are (tauto)logically equivalent, or whether one is the consequence of another.

Checking consequences, equivalence

Suppose you are asked to check whether two sentences are (tauto)logically equivalent, or whether one is the consequence of another.

- Logical consequence:
  $P_1, \ldots, P_n \vdash Q$ iff $P_1 \wedge \cdots \wedge P_n \rightarrow Q$ is logical truth
- Tautological consequence:
  $P_1, \ldots, P_n \Vdash Q$ iff $P_1 \wedge \cdots \wedge P_n \rightarrow Q$ is tautology
- Logical equivalence:
  $P \Leftrightarrow Q$ iff $P \leftrightarrow Q$ is logical truth
- Tautological equivalence:
  $P \equiv Q$ iff $P \leftrightarrow Q$ is tautology

Today's Lecture

1 Overview

2 Proof practice

3 Conversational implicature

4 Proofs with (bi)conditionals

5 Truth tables, tautologies and logical truths

6 Concluding

System $\mathcal{F}$

- Reiteration rule ✓
- Introduction and Elimination rules for identity ($=$) ✓
- Introduction and Elimination rules for $\wedge$, $\neg$, $\bot$, $\vee$ ✓
- Introduction and Elimination rules for conditionals $\rightarrow$, $\leftrightarrow$ ✓
- Introduction and Elimination rules for quantifiers

Wrapping up

Learning goals: At the end of this week, you are able to

- express English conditional statements in PL/FOL using (bi)conditionals.
- construct formal proofs, with nested subproofs, for sentences involving (bi)conditionals (and Boolean connectives).

Next week:

- More on tautologies vs logical truths, the truth table method, normal forms.